

CHICAGO MIDWAY, IL, USA

WMO: 725340

Lat: 41.786N

Lon: 87.752W

Elev: 187

StdP: 99.10

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14819

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.8	-15.0	-23.4	0.5	-16.5	-21.0	0.6	-14.2	11.6	-1.1	10.8	-1.8	5.0	270	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	33.2	23.7	32.0	22.9	30.5	22.2	25.5	31.1	24.4	29.6	23.6	28.5	5.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	19.0	28.9	22.7	17.8	27.7	22.0	17.1	27.0	79.0	31.2	74.6	29.7	71.1	28.4	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.4	8.5	-21.5	35.9	3.8	2.1	-24.3	37.4	-26.5	38.7	-28.6	39.9	-31.4	41.5		
(5)			-22.0	26.9	3.6	1.4	-24.6	27.9	-26.7	28.7	-28.8	29.5	-31.4	30.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	-3.6	-1.5	4.1	10.3	16.2	21.9	24.5	23.6	19.9	12.8	5.6	-0.5	(6)
(7)		DBStd	11.02	6.48	6.22	6.02	5.18	5.07	4.13	3.20	2.89	4.25	4.97	5.39	5.97	(7)
(8)		HDD10.0	1521	422	326	203	57	6	0	0	0	0	27	152	329	(8)
(9)		HDD18.3	3230	680	557	444	247	104	14	1	2	32	183	383	584	(9)
(10)		CDD10.0	1943	1	2	19	66	198	356	449	421	296	114	19	3	(10)
(11)		CDD18.3	611	0	0	2	6	39	120	191	164	78	12	0	0	(11)
(12)		CDH23.3	5188	0	0	11	57	351	1076	1768	1260	583	82	0	0	(12)
(13)		CDH26.7	1750	0	0	1	11	107	382	661	391	182	16	0	0	(13)
(14)	Wind	WSAvg	4.4	4.9	4.9	4.9	5.1	4.5	4.0	3.8	3.6	3.8	4.3	4.7	4.6	(14)
(15)	Precipitation	PrecAvg	908	51	47	59	85	103	104	97	101	73	76	63	49	(15)
(16)		PrecMax	1175	117	154	128	185	190	266	205	228	262	213	152	117	(16)
(17)		PrecMin	572	13	12	20	34	11	26	43	29	12	24	8	12	(17)
(18)		PrecStd	136	28	29	28	36	43	51	40	49	51	49	38	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.5	17.3	25.0	27.9	32.1	34.1	36.5	34.1	33.2	28.9	21.3	16.9	(19)
(20)			MCWB	12.7	11.9	17.3	17.6	21.3	23.9	25.8	24.2	22.9	20.2	14.5	15.0	(20)
(21)		2%	DB	10.3	13.0	20.3	24.8	29.5	32.4	33.7	32.2	31.0	25.8	18.0	12.9	(21)
(22)			MCWB	7.7	9.2	14.0	16.1	20.1	22.8	24.7	23.2	21.8	17.8	13.6	10.4	(22)
(23)		5%	DB	7.1	9.3	16.5	22.2	27.4	30.9	32.0	30.3	28.7	23.0	16.2	10.1	(23)
(24)			MCWB	5.0	6.4	11.8	14.5	18.9	22.4	23.5	22.6	20.5	16.6	12.4	7.8	(24)
(25)		10%	DB	4.7	6.4	13.0	18.9	24.9	28.8	30.2	28.8	27.0	20.9	13.7	7.4	(25)
(26)			MCWB	3.0	4.1	8.8	12.6	17.8	21.3	22.5	21.9	19.6	15.6	10.1	5.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	13.6	17.6	20.0	23.2	26.0	27.7	26.2	24.4	21.7	16.8	15.2	(27)
(28)			MCD B	14.4	15.9	24.2	26.0	29.6	31.9	33.4	32.4	30.6	27.4	18.9	16.5	(28)
(29)		2%	WB	8.3	9.9	15.1	17.3	21.5	24.5	25.8	24.9	23.0	19.7	14.9	11.2	(29)
(30)			MCD B	9.6	13.2	19.0	22.4	27.2	30.0	31.4	29.7	28.4	23.5	17.6	12.6	(30)
(31)		5%	WB	5.1	6.7	12.3	15.4	20.2	23.3	24.8	23.9	21.8	17.9	12.5	7.9	(31)
(32)			MCD B	6.8	9.1	16.1	21.0	25.3	28.6	30.2	28.2	26.6	21.6	15.6	9.5	(32)
(33)		10%	WB	3.0	4.2	9.1	13.3	18.8	22.3	23.8	23.0	20.7	16.2	10.5	5.2	(33)
(34)			MCD B	4.6	6.4	13.1	18.4	23.6	27.5	28.8	27.3	25.4	20.3	13.2	7.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.0	7.3	8.2	9.5	9.8	9.4	8.7	8.2	8.9	8.6	7.4	6.4	(35)
(36)			MCD BR	9.2	10.9	12.6	14.1	12.4	10.9	10.1	10.0	11.0	11.9	10.2	8.8	(36)
(37)			MCWBR	8.1	8.5	8.5	8.1	6.1	5.1	4.3	4.3	4.8	6.7	7.4	7.6	(37)
(38)		5% WB	MCD BR	8.9	10.0	11.8	12.8	11.2	10.1	9.5	8.6	9.3	9.9	9.5	8.2	(38)
(39)			MCWBR	8.2	8.7	8.9	8.8	6.7	5.6	5.0	4.5	4.9	7.1	8.0	7.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.303	0.322	0.352	0.407	0.436	0.441	0.452	0.448	0.418	0.364	0.321	0.303	(40)	
(41)		taud	2.434	2.373	2.370	2.237	2.211	2.243	2.212	2.219	2.268	2.419	2.495	2.472	(41)	
(42)		Ebn at Noon	846	882	894	863	843	837	824	814	812	823	820	815	(42)	
(43)		Edh at Noon	77	96	109	134	141	137	140	135	120	92	73	69	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.59	2.46	3.58	4.62	5.27	6.05	6.09	5.30	4.39	2.89	1.84	1.33	(44)	
(45)		RadStd	0.17	0.32	0.27	0.38	0.44	0.40	0.30	0.31	0.37	0.31	0.19	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page